

University of Cape Town
Department of Electrical Engineering
Class Test 2 - SOLUTION
9 October 2025

EEE2047S: Signals and Systems I

Instructions

Time: 90 minutes Full marks = 60 marks

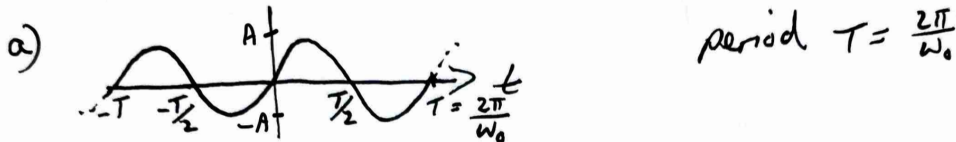
- Answer all questions.
- Write your answers on the question paper itself.
- Additional working may be written on the reverse sides (clearly indicate).
- Also provided: Information sheet with formulas.
- Write your EMPLID (PeopleSoft number) on the top right hand corner of this cover page.

Internal examiner: A/Prof A.J. Wilkinson

1. (12 marks) A voltage waveform is described by $v(t) = A \cos(\omega_0 t + \theta)$ with real constants $A > 0$, $\omega_0 > 0$ and $\theta = -\pi/2$.

- Plot $v(t)$ and label significant points on the axes. (2 marks)
- Determine the complex Fourier series and plot the magnitude and phase of the Fourier coefficients. (3 marks)
- Determine the Fourier transform $V(\omega)$ and draw the spectrum. (3 marks)
- Plot the instantaneous power as a function of time. (2 marks)
- Derive the average power of $v(t)$ making use of Parseval's theorem. (2 marks)

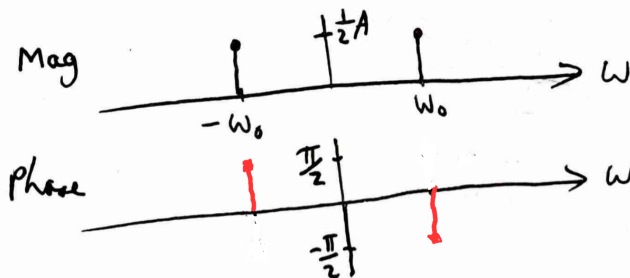
Q1 $v(t) = A \cos(\omega_0 t + \theta)$ $\theta = -\pi/2$ $A > 0$ $\omega_0 > 0$



b)
$$v(t) = \frac{1}{2} A e^{j(\omega_0 t + \theta)} + \frac{1}{2} A e^{-j(\omega_0 t + \theta)}$$

$$= \frac{1}{2} A e^{j\theta} e^{j\omega_0 t} + \frac{1}{2} A e^{-j\theta} e^{-j\omega_0 t}$$

$$\therefore c_1 = \frac{1}{2} A e^{j\theta} = \frac{1}{2} A e^{-j\pi/2} \quad c_{-1} = \frac{1}{2} A e^{-j\theta} = \frac{1}{2} A e^{j\pi/2}$$



c)
$$V(\omega) = \sum_{k=-\infty}^{\infty} 2\pi c_k \delta(\omega - k\omega_0) = 2\pi c_{-1} \delta(\omega + \omega_0) + 2\pi c_1 \delta(\omega - \omega_0)$$



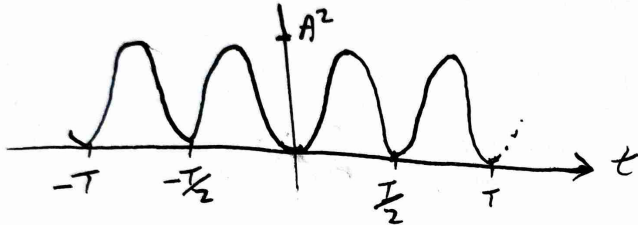
One can also express coefficients in rectangular form:

$$c_1 = \frac{1}{2} A e^{-j\pi/2} = -j \frac{A}{2}$$

$$c_{-1} = \frac{1}{2} A e^{j\pi/2} = j \frac{A}{2}$$

Note: Since $v(t)$ is a real function, $c_{-1} = c_1^*$

$$\begin{aligned}
 d) \quad p(t) &= |v(t)|^2 = A^2 \cos^2(\omega_0 t - \pi/2) = A^2 \sin^2(\omega_0 t) \\
 &= \frac{1}{2} A^2 - \frac{1}{2} A^2 \cos(2\omega_0 t)
 \end{aligned}$$



e) Parseval's Theorem for periodic signals

$$\frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2 = \text{Average power}$$

$$\begin{aligned}
 P_{ave} &= |c_{-1}|^2 + |c_1|^2 \\
 &= \left| \frac{1}{2} A e^{-j\theta} \right|^2 + \left| \frac{1}{2} A e^{j\theta} \right|^2 \\
 &= \frac{1}{4} A^2 + \frac{1}{4} A^2 \\
 &= \frac{1}{2} A^2
 \end{aligned}$$

Note: There are two different versions of Parseval's theorem, one for calculating average power from the Fourier series coefficients (relevant to periodic signals with a Fourier series), and the other for calculating energy from the Fourier transform (used for signals which have finite energy, like a top-hat pulse):

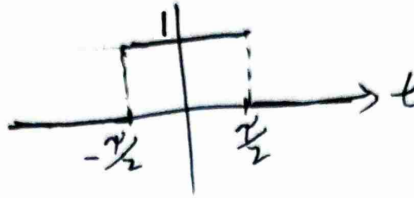
$$P_{ave} = \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2$$

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

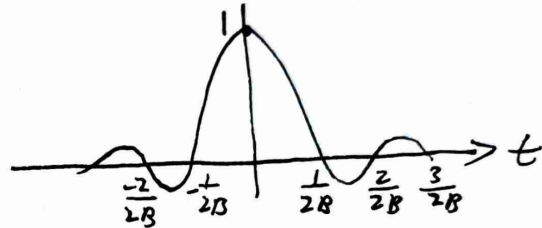
2. (12 marks) A signal $x(t) = p_\tau(t)$ is fed into a LTI system with impulse response $h(t) = \text{Sa}(2\pi Bt)$. The parameters $\tau > 0$ and $B > 0$ are real constants. Let the output signal be $y(t)$.

- Plot $x(t)$ (2 marks)
- Plot $h(t)$ (2 marks)
- Find $Y(\omega) = \mathcal{F}\{y(t)\}$ (5 marks)
- Plot $Y(\omega)$ for the case where $B = 2/\tau$ labelling significant features. (3 marks)

a) $x(t) = p_\tau(t)$



b) $h(t) = \text{Sa}(2\pi Bt)$ $B > 0$



Zero-crossings
when $2\pi Bt = n\pi$
 $\therefore t = n \frac{1}{2B}$

c) $Y(\omega) = X(\omega) H(\omega)$ where $X(\omega) = \tau \text{Sa}(\frac{\gamma\omega}{2})$

from table $\tau \text{Sa}(\frac{\gamma t}{2}) \leftrightarrow 2\pi p_\tau(\omega)$

$\therefore \text{Sa}(\frac{\gamma}{2}t) \leftrightarrow \frac{2\pi}{\gamma} p_\tau(\omega)$

$\mathcal{F}\{\text{Sa}(2\pi Bt)\} = \mathcal{F}\{\text{Sa}(\frac{\gamma}{2}t)\}$ with $2\pi B = \frac{\gamma}{2}$
 $\therefore \gamma = 4\pi B$

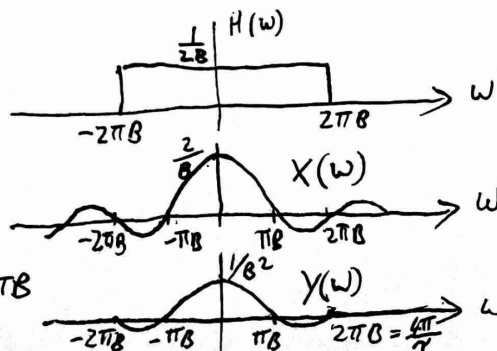
$= \frac{2\pi}{4\pi B} p_{4\pi B}(\omega) = \frac{1}{2B} p_{4\pi B}(\omega)$

$\therefore Y(\omega) = \tau \text{Sa}(\frac{\gamma\omega}{2}) \cdot \frac{1}{2B} p_{4\pi B}(\omega)$

d) Plot $Y(\omega)$ for $B = 2/\tau$

$X(\omega) = \tau \text{Sa}(\frac{\gamma\omega}{2})$
 $= \frac{2}{B} \text{Sa}(\frac{\omega}{B})$

zero crossings at $\frac{\omega}{B} = n\pi \Rightarrow \omega = n\pi B$



3. (10 marks) A signal $x(t)$ has Fourier transform $X(\omega)$.
Express the Fourier transforms of the functions below in terms of $X(\omega)$:

(a) $x_1(t) = x(1-t) + x(-2-t)$ (5 marks)

(b) $x_2(t) = x(2t-4)$ (5 marks)

(a) Applying the time reversal property to the pair $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$ gives

$$x(-t) \xleftrightarrow{\mathcal{F}} X(-\omega)$$

Using the time-shift property on this pair gives

$$x(-(t-c)) \xleftrightarrow{\mathcal{F}} X(-\omega)e^{-j\omega c}.$$

Using $c = 1$ and $c = -2$ in this pair gives the required result:

$$\begin{aligned} X_1(\omega) &= X(-\omega)e^{-j\omega} + X(-\omega)e^{j2\omega} \\ &= X(-\omega)(e^{-j\omega} + e^{j2\omega}). \end{aligned}$$

(b) Applying the time scaling property to the pair $x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$ gives

$$x(2t) \xleftrightarrow{\mathcal{F}} \frac{1}{2}X\left(\frac{\omega}{2}\right).$$

Time shifting on this pair yields

$$x(2(t-c)) \xleftrightarrow{\mathcal{F}} \frac{1}{2}X\left(\frac{\omega}{2}\right)e^{-j\omega c}.$$

The case of $c = 2$ gives the required result:

$$X_2(\omega) = \frac{1}{2}X\left(\frac{\omega}{2}\right)e^{-j2\omega}.$$

Note: Another way to do part (b) is to write

$$\begin{aligned} x_2(t) &= x(2t-4) \\ &= x(2(t-2)) \\ &= g(t-2) \end{aligned}$$

where

$$g(t) = x(2t)$$

Then (from the time shift property)

$$X_2(\omega) = \mathcal{F}\{g(t-2)\} = G(\omega)e^{-j2\omega}$$

where $G(\omega)$ is obtained by applying the time scaling property:

$$G(\omega) = \mathcal{F}\{x(2t)\} = \frac{1}{2}X\left(\frac{\omega}{2}\right)$$

Combining

$$X_2(\omega) = \frac{1}{2}X\left(\frac{\omega}{2}\right)e^{-j2\omega}$$

4. (5 marks) Use tables and properties to find

$$\mathcal{F}\left\{\frac{1}{5+jt}\right\}$$

Q4

$$\mathcal{F}\left\{\frac{1}{5+jt}\right\}$$

From table

$$e^{-bt}u(t) \leftrightarrow \frac{1}{b+j\omega} \quad (b>0)$$

Apply Duality Property $X(t) \leftrightarrow 2\pi x(-\omega)$

$$\therefore \frac{1}{b+jt} \leftrightarrow 2\pi e^{b\omega}u(\omega)$$

$$\therefore \mathcal{F}\left\{\frac{1}{5+jt}\right\} = 2\pi e^{5\omega}u(-\omega)$$



5. (13 marks) A system is described by the following differential equation

$$\frac{d}{dt}y(t) + y(t) = \frac{d}{dt}x(t) - x(t).$$

- (a) Derive the transfer function $H(\omega)$ (3 marks)
 (b) Determine and plot the impulse response $h(t)$ (7 marks)
 (c) Find the step response. (3 marks)

(a) Take Fourier transform of left and right sides:

$$j\omega Y(\omega) + Y(\omega) = j\omega X(\omega) - X(\omega)$$

$$Y(\omega)(j\omega + 1) = X(\omega)(j\omega - 1)$$

we have

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{j\omega - 1}{j\omega + 1}.$$

(b) To obtain inverse transform, split into two terms:

$$H(\omega) = j\omega \left(\frac{1}{j\omega + 1} \right) - \left(\frac{1}{j\omega + 1} \right)$$

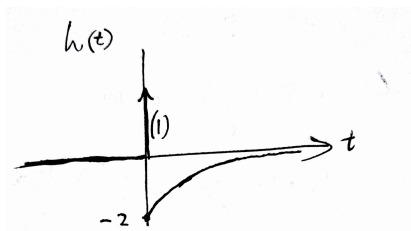
From tables, $e^{-t}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{j\omega + 1}$.

Apply the time differentiation property $\frac{d^n}{dt^n}x(t) \longleftrightarrow (j\omega)^n X(\omega)$ $n = 1, 2, \dots$ to get the pair

$$\frac{d}{dt}e^{-t}u(t) \xleftrightarrow{\mathcal{F}} \frac{j\omega}{j\omega + 1},$$

so the impulse response is

$$h(t) = \frac{d}{dt} \{e^{-t}u(t)\} - e^{-t}u(t) = \delta(t) - e^{-t}u(t) - e^{-t}u(t) = \delta(t) - 2e^{-t}u(t).$$

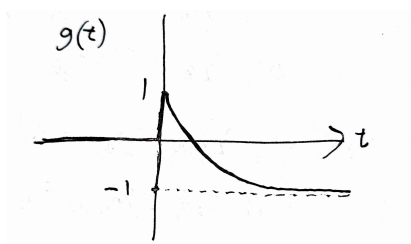


(c) The step response is $g(t) = \int_{\tau=-\infty}^{\infty} h(\tau)u(t-\tau)d\tau = \int_{\tau=-\infty}^t h(\tau)d\tau = \int_{-\infty}^t [\delta(\tau) - 2e^{-2\tau}u(\tau)] d\tau$.
 For $t < 0$ we have $g(t) = 0$, and for $t \geq 0$

$$\begin{aligned} g(t) &= u(t) - 2 \int_{\tau=0}^t e^{-\tau} d\tau \\ &= 1 - 2 [-e^{-\tau}]_{\tau=0}^t = 1 - 2(1 - e^{-t}) = 2e^{-t} - 1 \end{aligned}$$

so

$$g(t) = u(t) (2e^{-t} - 1)$$



Another (more tedious) approach to obtain the step response is via $g(t) = \mathcal{F}^{-1} \{ \mathcal{F} \{ u(t) \} H(\omega) \} = \text{etc.}$

To obtain the step response is via $g(t) = \mathcal{F}^{-1} \{ \mathcal{F} \{ u(t) \} H(\omega) \}$ is significantly more tedious:

$$\begin{aligned}
 G(\omega) &= \mathcal{F} \{ u(t) \} H(\omega) \\
 &= \left(\pi \delta(\omega) + \frac{1}{j\omega} \right) \left(\frac{j\omega - 1}{j\omega + 1} \right) \\
 &= \pi \delta(\omega) \left(\frac{j\omega - 1}{j\omega + 1} \right) + \frac{1}{j\omega} \left(\frac{j\omega - 1}{j\omega + 1} \right) \\
 &= \pi \delta(\omega) \left(\frac{j0 - 1}{j0 + 1} \right) + \frac{1}{j\omega} \left(\frac{j\omega - 1}{j\omega + 1} \right) \\
 &= -\pi \delta(\omega) + \left(\frac{1}{j\omega + 1} \right) - \frac{1}{j\omega} \left(\frac{1}{j\omega + 1} \right)
 \end{aligned}$$

Apply partial fractions decomposition to the last term:

$$\frac{1}{j\omega} \left(\frac{1}{j\omega + 1} \right) = \frac{A}{j\omega} - \left(\frac{B}{j\omega + 1} \right) = \frac{1}{j\omega} - \left(\frac{1}{j\omega + 1} \right)$$

Thus

$$\begin{aligned}
 G(\omega) &= -\pi \delta(\omega) + \left(\frac{1}{j\omega + 1} \right) - \left[\frac{1}{j\omega} - \left(\frac{1}{j\omega + 1} \right) \right] \\
 &= - \left[\pi \delta(\omega) + \frac{1}{j\omega} \right] + 2 \left(\frac{1}{j\omega + 1} \right)
 \end{aligned}$$

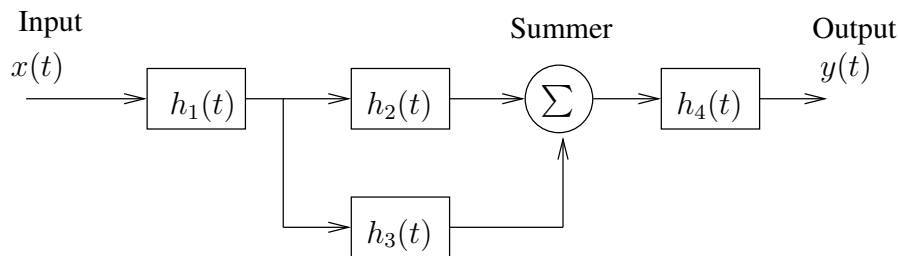
Taking the inverse transform using tables gives the step response:

$$\begin{aligned}
 g(t) &= -u(t) + 2e^{-t}u(t) \\
 &= u(t) (2e^{-t} - 1)
 \end{aligned}$$

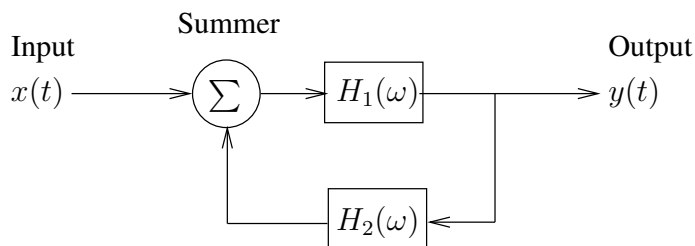
which agrees with the other method.

6. (8 marks) Block diagrams:

(a) Derive the system transfer function $H(\omega)$ in terms of the parameters given. (4 marks)



(b) Derive the transfer function $H(\omega)$ for the following feedback system: (4 marks)

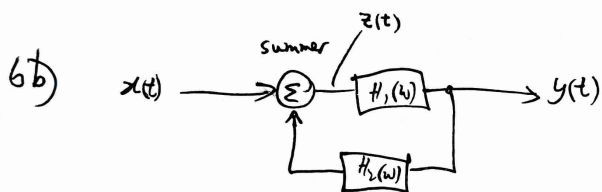


a) Let $H_i(\omega) = \mathcal{F}\{h_i(t)\}$

$$Y(\omega) = H_4(\omega) \cdot [X(\omega) H_1(\omega) H_2(\omega) + X(\omega) H_1(\omega) H_3(\omega)]$$

$$= X(\omega) H_1(\omega) H_4(\omega) [H_2(\omega) + H_3(\omega)]$$

$$\therefore H(\omega) = \frac{Y(\omega)}{X(\omega)} = H_1(\omega) H_4(\omega) [H_2(\omega) + H_3(\omega)]$$



$$Z(\omega) = X(\omega) + H_2(\omega) Y(\omega)$$

$$Y(\omega) = H_1(\omega) Z(\omega)$$

$$\therefore Y(\omega) = H_1(\omega) X(\omega) + H_1(\omega) H_2(\omega) Y(\omega)$$

$$\therefore Y(\omega) [1 - H_1(\omega) H_2(\omega)] = H_1(\omega) X(\omega)$$

$$\therefore H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{H_1(\omega)}{1 - H_1(\omega) H_2(\omega)}$$